# CH4_Inventory_Main.R

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`CH4_inventory_build` *Main function to calculate gridded methane emissions. Runs sector by sector according to the config file.*

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### Description

'CH4_inventory_build' runs multiple other functions to calculate the gridded methane emissions sector-by-sector using information provided in the config file and inputs.

### Usage

```
CH4_inventory_build(
  input_directory,
  output_directory,
  code_directory,
  plot_directory,
  inventory_year,
  domain = as.data.frame(cbind(c(-76.65, -73.65), c(38.97, 40.97))),
  domain_res = 0.01,
  domain_crs = "epsg:4326",
  ACES_directory,
  vulcan_directory,
  verbose,
  EIA_file,
  HIFLD_compressor_file,
  NLCD_file,
  NALCMS_file,
  DMR_file,
  CWNS_file,
  watershed_shapefile
)
```

### Arguments

`input_directory`
: Character providing the full filepath to load/save input data

`code_directory`
: Character providing the full filepath to source all other functions and the config.

`plot_directory`
: Character providing the full filepath to save figures. Only relevant if verbose = TRUE.

`inventory_year`
: Numeric indicating the desired year of data to use. The closest available will be used for datasets that are not provided annually.

`domain`
: data.frame or character. If data.frame, provides the corner coordinates of the desired region to process. The first column would be x values, the second column y values, and the first row would be the minima, the second row the maxima. These must be provided in the appropriate units for the domain_crs parameter (e.g., decimal degrees for a lat/long projection). If character, it indicates the desired region using census Tigerlines. It can be a state abbreviation, state name, state FIPS code (as a character), Urban Area name, or Urban Area Census Code. For example: "DE", "Delaware", and "10" are all equivalent and "Long Neck, DE" and "51202" are equivalent. Names must match exactly and FIPS or urban area codes must include leading 0's. Lists of the names and codes for all urban areas for the most recent census are available at https://www.census.gov/programs-surveys/geography/guidance/geo-areas/urban-rural.html. Links to previous census urban areas are on the same page. A list with state fips codes is available here https://www.census.gov/library/reference/code-lists/ansi.html.

`domain_res`
: Numeric providing the resolution for the domain. Can be length 1 for equal x and y resolution or length 2 (x, y).

`domain_crs`
: Character providing the projection of the domain in PROJ string (https://proj.org/en/stable/operations/projections/index.html) or EPSG codes (https://epsg.io/) or WKT (https://docs.ogc.org/is/12-063r5/12-063r5.html) formats.

`ACES_directory`
: Character providing the full filepath to a folder with sectoral Anthropogenic Carbon Emission System (ACES) inventory files that have been averaged into annual files. ACES is available at https://doi.org/10.3334/ORNLDAAC/1943.

`vulcan_directory`
: Character providing the full filepath to a folder with annual, sectoral Vulcan inventory files. Vulcan v3.0 is available at https://doi.org/10.3334/ORNLDAAC/1741.

`verbose`
: Logical indicating whether to save additional output. This includes easier to read csv files of the data that has been gridded for multiple sectors as well as sector and subsector visuals.

`EIA_file`
: Character providing the full filepath to the EIA form 176 data in xlsx format available at https://www.eia.gov/naturalgas/ngqs/#?report=RP4&year1=2020&year2=2020&company=Name.

`HIFLD_compressor_file`
: Character providing the full filepath to the HIFLD natural gas transmission compressor data in csv format. This data has been deprecated and is no longer available from HIFLD.

`NALCMS_file`
: Character providing the full filepath to the NALCMS landcover data in tif format available at http://www.cec.org/north-american-land-change-monitoring-system/.

`DMR_file` Character providing the full filepath to the DMR wastewater treatment plant data in csv format available at https://echo.epa.gov/trends/loading-tool/water-pollution-search.

`CWNS_file` Character providing the full filepath to the CWNS wastewater treatment plant data. For 2012 this should be in xlsx format and for 2022 this should be a folder with multiple csv's. The 2012 CWNS is available at https://ordspub.epa.gov/ords/cwns2012/f?p=cwns2012:25: and the 2022 CWNS is available at https://sdwis.epa.gov/ords/sfdw_pub/r/sfdw/cwns_pub/data-download?session=8491118346746.

`watershed_shapefile` Character providing the full filepath to the watershed polygons in shapefile format available at http://www.cec.org/north-american-environmental-atlas/watersheds/.

## Details

This function will source all other functions and the config, create the domain from the input data, and download some files (Census tigerlines, ghgrp location/name data) that will be needed for multiple sectors. It will then use the information in the config file and inputs to run every relevant sector. As such, the inputs to this function and the config may require user editing.

See references Vulcan and ACES

## Value

Nothing is returned from the function, but there will be frequent user updates as multiple other functions are run and output files created for the various sectors.

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## Examples

```
CH4_inventory_build(input_directory="G:/My Drive/Shepson Group Drive/Kris/Philly Inventory/Raw_data_rewrite/
             output_directory="G:/My Drive/Shepson Group Drive/Kris/Philly Inventory/Processed_rewrite/",
                    code_directory="~/../../Kristian/Desktop/methane_inventory/src/",
             plot_directory="G:/My Drive/Shepson Group Drive/Kris/Philly Inventory/Figures_rewrite/",
                    inventory_year=2019,
                    domain=as.data.frame(cbind(c(-76.65,-73.65),
                                               c(38.97,40.97))),
                    domain_res=0.01,
                    domain_crs="epsg:4326",
             NALCMS_file = "G:/My Drive/Shepson Group Drive/Kris/Philly Inventory/Large inventory files/NA_NA
             NLCD_file = "G:/My Drive/Shepson Group Drive/Kris/Philly Inventory/Large inventory files/nlcd_20
             ACES_directory="G:/My Drive/Shepson Group Drive/Kris/Philly Inventory/Large inventory files/ACE
             vulcan_directory="G:/My Drive/Shepson Group Drive/Kris/Philly Inventory/Large inventory files/V
             DMR_file = "~/../../Kristian/Desktop/methane_inventory/src/Data/DMR_2019_from_11_1_2024.csv",
             CWNS_file = "~/../../Kristian/Desktop/methane_inventory/src/Data/CWNS_merged_data_2012_KH.xlsx
             EIA_file = "~/../../Kristian/Desktop/methane_inventory/src/Data/EIA_company_report_2019.xlsx",
             HIFLD_compressor_file="~/../../Kristian/Desktop/methane_inventory/src/Data/Natural_Gas_Compre
             watershed_shapefile="~/../../Kristian/Desktop/methane_inventory/src/Data/watersheds_shapefile
             ACES_directory="G:/My Drive/Shepson Group Drive/General Inventories and Shapefiles/Inventories/
```

```
vulcan_directory="G:/My Drive/Shepson Group Drive/General Inventories and Shapefiles/Inventorie
          verbose=TRUE)
```

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